

Advanced Algebra



Simplifying rational expressions

- 1 Simplify $\frac{x^2 - 4}{x^2 + x - 6}$.
 - 2 Simplify $\frac{x^2 - 5x + 6}{x^2 - 9}$.
 - 3 Combine into a single fraction: $\frac{2}{x+1} + \frac{3}{x-1}$.
 - 4 Simplify $\frac{3x^2 - 12}{x^2 - 4x + 4}$.
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Absolute value equations

- 5 Solve $|2x - 5| = 9$.
 - 6 Solve $|3x + 1| = |x - 5|$.
 - 7 Solve $|x - 4| + 3 = 10$.
 - 8 Solve $|4x + 3| = 11$.
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Polynomial division, remainder & factor theorems

- 9 Use the remainder theorem to find the remainder when $f(x) = x^3 + 2x^2 - 5x + 1$ is divided by $(x - 2)$.
 - 10 Given $f(x) = x^3 + 3x^2 - 4x - 12$:
 - a Use the factor theorem to show that $(x + 3)$ is a factor of $f(x)$.
 - b Hence factor $f(x)$ completely.
 - 11 Divide $x^3 - 2x^2 - 5x + 6$ by $(x - 3)$ using polynomial long division. State the quotient and remainder.
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Logarithmic equations

- 12 Solve $\log_3(x + 5) = 2$.
- 13 Solve $\log_2(x + 1) + \log_2(x - 1) = 3$.

14 Solve $2\log_5 x = \log_5 16$.

15 Solve $\log_3 (x - 2) = 4$.

Exponential equations

16 Solve $3^{2x-1} = 27$.

17 Solve $5^{x+2} = 5^{3x-4}$.

18 Solve $2^x = 15$ for x , giving your answer to three decimal places.

19 Solve $4^x = 2^{x+3}$.

Nonlinear systems of equations

20 Solve the system $y = x^2 - 1$ and $y = 2x + 2$.

21 Solve the system $x^2 + y^2 = 25$ and $y = x + 1$.

22 Solve the system $y = x^2 - 4x + 3$ and $y = -x + 3$.

Rational equations and extraneous solutions

23 Solve $\frac{4}{x-1} = \frac{2}{x+1}$.

24 Solve $\frac{x}{x-2} + \frac{2}{x+2} = \frac{8}{x^2-4}$.

25 Solve $\frac{3}{x+4} = \frac{1}{x-2}$.

Systems of inequalities and linear programming

26 A feasible region is bounded by the constraints $x \geq 0$, $y \geq 0$, $x + y \leq 6$, and $2x + y \leq 10$. Find the vertices of the feasible region.

27 Using the feasible region from the previous question, find the maximum value of $P = 3x + 2y$ over the region, and state where it occurs.

A well-designed word problem: production planning at a furniture workshop

- 28** This question has three parts. A furniture workshop makes tables and chairs. Each table requires 4 hours of carpentry and 2 hours of finishing; each chair requires 1 hour of carpentry and 1 hour of finishing. Each week the shop has at most 32 hours of carpentry time and 20 hours of finishing time available. Each table earns a profit of 70 dollars and each chair earns a profit of 30 dollars. Let x be the number of tables and y the number of chairs produced in a week. (a) Write the system of inequalities describing the constraints (including $x \geq 0$ and $y \geq 0$). (b) Find the vertices of the feasible region. (c) Determine the number of tables and chairs that maximizes the weekly profit $P = 70x + 30y$, and find the maximum profit.